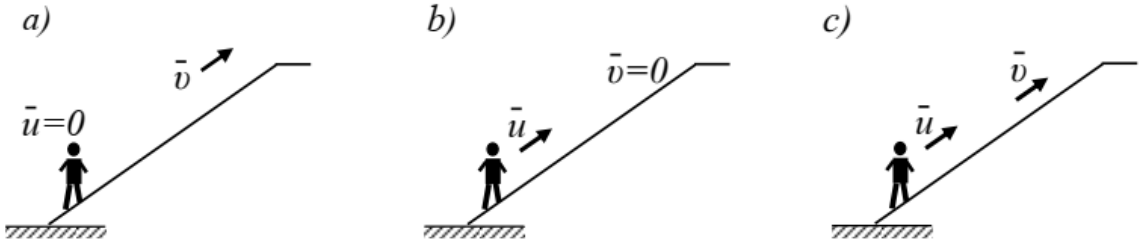
Fizika

1. Metro eskalatori tinch turgan yo'lovchini $t_1 = 1$ minutda olib chiqadi. Eskalator to'xtab turganda odam unda harakatlanib $t_2 = 3$ minutda ko'tariladi. Harakatdagi eskalatorda odam yurib qancha vaqtda ko'tariladi? Har bir holda eskalator va odamning tezligi o'zgarmas.

Berilgan: $t_1 = 60s$, $t_2 = 180\,s$;
Topish kerak: $t_3 - ?$



Masalaning yechilishi:

Eskalator uzunligini l bilan, uning harakat tezligini υ, harakatlanayotgan odamning eskalatorga nisbatan tezligini u bilan belgilaymiz. Harakatni eskalatorning qo'zg'almas korpusi bilan bog'laymiz. Masala shartidagi holatlar mos ravishda 1.5–rasm (*a, b, c*) da keltirilgan. To'g'ri chiziqli tekis harakat ifodasiga asosan quyidagi tenglamalar o'rinli:

$$l = \upsilon t_1 \qquad (1)$$
$$l = u t_2 \qquad (2)$$
$$l = (\upsilon + u) t_3 \qquad (3)$$

(1)–(3) tenglamalarni birgalikda yechib, masalaning ishchi formulasini olamiz:

$$\boxed{t_3 = \frac{t_1 t_2}{t_1 + t_2}} \qquad (4)$$

$$[t] = \frac{s^2}{s} = s;$$

Hisoblash natijasi: $t_3 = 45\ s$.

2. 5 m uzunlikdagi arqonga osilgan 45 kg massali jism 16,0 ayl/min chastota bilan aylanmoqda. Arqon vertikal bilan qanday burchak tashkil qiladi va uning taranglik kuchi qanday?

Berilgan: $m = 45\ kg,\ l = 5\ m,\ \nu = 0{,}27\ s^{-1},\ g = 9{,}8\ m/s^2;$
Topish kerak: $\alpha - ?\quad T - ?$
Masalaning yechilishi:

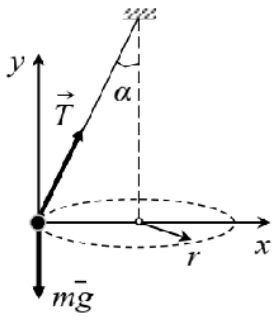
Jismga og'irlik kuchi hamda ipning taranglik kuchi ta'sir etadi. Jism harakati uchun dinamikaning asosiy tenglamasini vektor ko'rinishida yozamiz:

$$m\bar{g} + \bar{T} = m\bar{a} \qquad (1)$$

Ox o'qini gorizontal, *Oy* o'qini unga tik yo'naltirib, (1) ifodaga kirgan vektor kattaliklarni mazkur o'qlarga proeksiyalarini olamiz:

Ox o'qiga: $T\sin\alpha = ma$ (2)
Oy o'qiga: $-mg + T\cos\alpha = 0$ (3)

Markazga intilma tezlanish ifodasi $a = 4\pi^2\nu^2 r$, bu yerda *r*–jism harakatlanayotgan aylana radiusi. Rasmdan $r = l\sin\alpha$. Bu ifodalarni e'tiborga olib, (2) va (3) dan izlanilayotgan kattaliklarni hisoblash tenglamalarini yozish mumkin:



$$\boxed{\alpha = \arccos(g/4\pi^2\nu^2 l)} \qquad (4)$$

$$\boxed{T = 4\pi^2\nu^2 l\, m} \qquad (5)$$

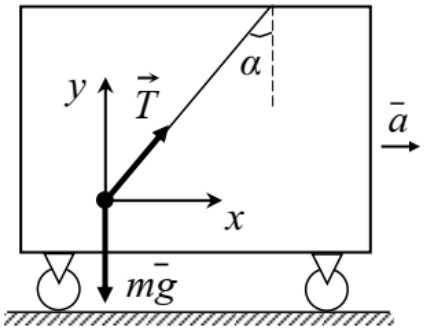
T ning o'lchov birligini tekshiramiz: $[T] = \dfrac{kg\cdot m}{s^2} = N;$

Hisoblash natijalari: $\alpha = 45^0,\ T = 0{,}63\ kN.$

3. Gorizontal yo'nalishda doimiy 2 m/s^2 tezlanish bilan harakatlanayotgan vagon shipiga ip orqali 100 g massali jism osilgan. Agar yuk vagonga nisbatan tinch turgan bo'lsa, ipning vertikal bilan hosil qilgan burchagi va ipning taranglik kuchini toping.

Berilgan: $m = 0{,}1\ kg,\ a = 2\ m/s^2,\ g = 9{,}8\ m/s^2$;
Topish kerak: $\alpha - ?\quad T - ?$
Masalaning yechilishi:

Jismga og'irlik hamda ipning taranglik kuchi ta'sir etadi. Agar yuk qo'zg'almas bo'lsa, uning tezlanishi vagon tezlanishiga teng bo'ladi, natijada ip vertikaldan vagon harakat yo'nalishiga teskari tomonga og'adi Jism harakati uchun dinamikaning asosiy tenglamasini vektor ko'rinishida yozamiz:



$$m\bar{g} + \bar{T} = m\bar{a} \qquad (1)$$

Ox o'qini gorizontal, *Oy* o'qini unga tik yo'naltiramiz. (1) ifodaga kirgan vektor kattaliklarni mazkur o'qlarga proeksiyalarini olib, uni qayta yozamiz:

Ox o'qiga: $T\sin\alpha = ma$ (2)
Oy o'qiga: $-mg + T\cos\alpha = 0$ (3)

(2) va (3) tenglamalar nisbatidan ipni vertikaldan og'ish burchagini hisoblash formulasi kelib chiqadi:

$$\boxed{\alpha = arctg\left(\frac{a}{g}\right)} \qquad (4)$$

(2) va (3) tenglamalarni hadma–had qo'shish orqali masalaning ikkinchi sharti uchun ishchi formulaga ega bo'lamiz:

$$\boxed{T = m\sqrt{g^2 + a^2}} \qquad (5)$$

Hisoblash natijalari: $\alpha = 27^0,\ T = \sqrt{1{,}04}\ N.$

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